MEMORIAL UNIVERSITY

Department of Mathematics and Statistics
Faculty of Science

Scott MacLachlan, Professor

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To whom it may concern:

I am very pleased to write this letter for Federico Danieli, a Ph.D. student in the "Industrially Focused Mathematical Modelling" Centre for Doctoral Training at the University of Oxford, who is applying for a postdoctoral position in your group or department. I first met Federico in 2017, at a workshop in Switzerland on Parallel-in-Time integration techniques, when he was relatively early on in his Ph.D. studies. I then hosted Federico as a visiting graduate student for a total of 6 months in 2019, during which time I became much more familiar with his skills, interests, and abilities.

Federico's Ph.D. project was the outgrowth of a collaboration started in a "mini-project" with the Culham Centre for Fusion Energy (CCFE) on applying parallel-in-time algorithms to the modelling of plasmas in fusion reactors. Classical high-performance computing algorithms for the solution of time-dependent PDEs are commonly based on the idea of time-stepping, where a problem is discretized in space, and an approximation to the solution evolved in steps from an initial condition to some physically meaningful final time. Parallel-in-time algorithms aim to overcome a computational bottleneck in such approaches on large supercomputers, by solving for the approximate solution at all time points simultaneously, typically via some form of fixed-point iteration to approximately solve the nonlinear equations that define the solution of a discretized partial differential equation. While such parallel-in-time algorithms were first proposed over 50 years ago, and have been extensively studied for the past 20 years, many challenges remain in their use in practice, particularly for hyperbolic PDEs, such as those that model plasmas in certain flow configurations.

Federico's Ph.D. supervisor at Oxford, Prof. Andrew Wathen, has a long history of influential work on the development and analysis of numerical simulation algorithms, particularly for fluid flow, but has only recently turned his attention to parallel-in-time algorithms. In contrast, I have worked on parallel-in-time algorithms for the past 9 years, with a particular focus in recent years on developing prototype algorithms for the hyperbolic case. Thus, I was happy to oblige when Prof. Wathen suggested that Federico come to Memorial as a visiting graduate student. Federico visited me twice in 2019, for a total period of about six months. During this time we worked closely together, developing and testing some ideas for extensions to my recent work on scalar hyperbolic PDEs. The result of this work was a manuscript that we originally submitted for publication late in 2019 that is currently under review.

At the time he came to Memorial, Federico was a well-trained third-year Ph.D. student, and I found he had an excellent background in his research area but needed a little guidance to narrow his focus to tractable research questions. This was not particularly surprising to me: the equations that model plasmas are incredibly complex, combining aspects of the

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Navier-Stokes (or Euler) equations with Maxwell's Equations to model interactions between the flow of the plasma and the electromagnetic fields that act on it. The central research question proposed by his collaborators at CCFE was a difficult one, and is at least 5-10 years beyond the current state-of-the-art. Defining a likely path from where we are now in the field to where we would need to be to answer that question is a task far beyond what we would expect from a Ph.D. student, even an excellent one like Federico. I found, however, that he was very receptive to my guidance on tractable next steps, and am very happy to see that he is now producing excellent quality research in a number of promising directions.

Overall, I found Federico to be a well-motivated and extremely capable junior researcher in the area of Scientific Computation, who was equally well able to understand technical analysis of the discretization schemes that we considered and to implement practical calculations. He integrated seamlessly into my research group during his time at Memorial, even taking on the role of organizing my group seminar series for one of the semesters that he was here. I found Federico to be a hard worker, who worked well both by himself and in a team environment. I strongly recommend him to you for a postdoctoral position. Please do not hesitate to contact me if you require any further information on this recommendation.

Sincerely,

Scott MacLachlan
Professor
Department of Mathematics and Statistics
Memorial University of Newfoundland